
EECS 16A Designing Information Devices and Systems I

Discussion 5A

1. Proof

Concept: Null Spaces, Invertibility

Consider a square matrix $\mathbf{A}$. Prove that if $\mathbf{A}$ has a non-trivial nullspace, i.e. if the nullspace of $\mathbf{A}$ contains more than just $\vec{0}$, then the matrix $\mathbf{A}$ is not invertible.

2. Steady State Reservoir Levels

We have 3 reservoirs: A, B and C . The pumps system between the reservoirs is depicted in the figure below.

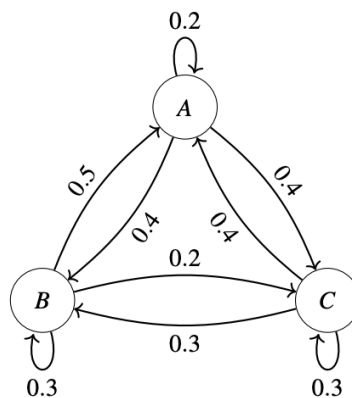


Figure 1: Reservoir pumps system.

- (a) Write out the transition matrix $\mathbf{T}$ representing the pumps system.

- (b) You are told that $\lambda_1 = 1$, $\lambda_2 = \frac{-\sqrt{2}-1}{10}$, $\lambda_3 = \frac{\sqrt{2}-1}{10}$ are the eigenvalues of $\mathbf{T}$. Find a steady state vector $\vec{x}$, i.e. a nonzero vector such that $T\vec{x} = \vec{x}$.

3. Row Operations and Determinants

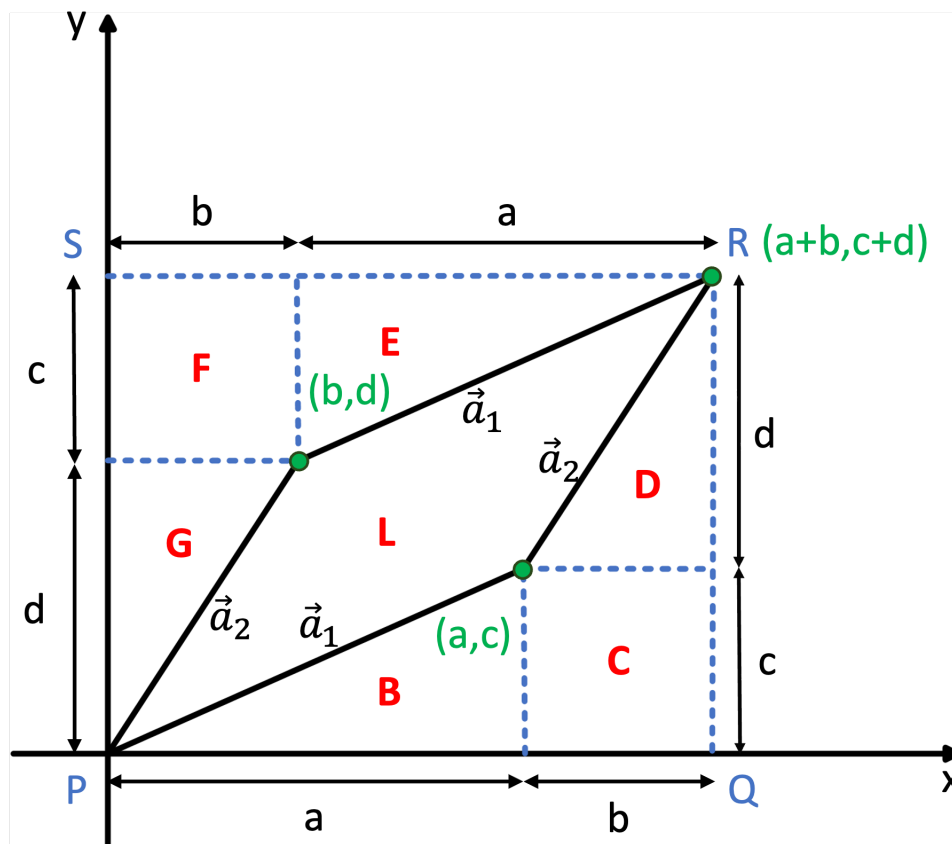


Figure 1: Parallelogram formed by columns of $\mathbf{A}$

Recall from lecture the way to compute a determinant of any 2×2 matrix is by using the following formula:

$$\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \det(\mathbf{A}) = ad - bc$$

There is a geometric interpretation of the determinant of a 2×2 matrix. The determinant of a 2×2 matrix $\mathbf{A}$ is the area of the parallelogram formed by the columns of $\mathbf{A}$. This generalizes to higher dimensions as well. For example, the determinant of a 3×3 matrix is the volume of the parallelepiped formed by the columns of the 3×3 matrix. In general, the determinant of an $n \times n$ matrix is the n -dimensional volume formed by the columns of that matrix.

We will show that this geometric relationship holds in the case of a 2×2 matrix. Consider the following matrix:

$$\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

This matrix has columns:

$$\vec{a}_1 = \begin{bmatrix} a \\ c \end{bmatrix}$$

$$\vec{a}_2 = \begin{bmatrix} b \\ d \end{bmatrix}$$

Figure 1 shows the parallelogram formed by the columns of $\mathbf{A}$.

To find the area of this parallelogram, we can find the area of the rectangle PQRS and subtract from this the areas of the triangles and smaller rectangles that are not part of the parallelogram. We can see from the figure that the area of rectangle PQRS is $(a+b)(c+d)$. Expanding, we get $\text{area}(\text{PQRS}) = ac + ad + bc + bd$. We'll call this area H , and we want to find the area denoted by L in the figure. We can see that $L = H - B - C - D - E - F - G$. Also, we can use geometry to find each area in terms of the entries of $\mathbf{A}$.

$$H = ac + ad + bc + bd$$

$$B = \frac{1}{2}ac$$

$$C = bc$$

$$D = \frac{1}{2}bd$$

$$E = \frac{1}{2}ac$$

$$F = bc$$

$$G = \frac{1}{2}bd$$

Substituting these into $L = H - B - C - D - E - F - G$, we get:

$$L = ac + ad + bc + bd - \frac{1}{2}ac - bc - \frac{1}{2}bd - \frac{1}{2}ac - bc - \frac{1}{2}bd = ad - bc$$

So, we have that the area of the parallelogram formed by the columns of $\mathbf{A}$ is $ad - bc = \det(\mathbf{A})$.

Compute the determinant of the following matrix using the formula above:

(a) $\begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$